

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester VI Examination, 2013-14

Paper No. BPT-603: Elements of Nuclear Physics

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer all questions. Always draw necessary diagram or figure if needed. Symbols have their usual meanings.

Question 1. Answer any **five** questions of the following: [5 × 3 = 15]

- (a) Nucleon-nucleon forces are charge independent. Why do the $p - p$ and $n - n$ bound states not exist?
- (b) On what factors does the choice of a good detector scintillator depend?
- (c) Explain, with one example each, the following terms: (i) Mirror Nuclei, (ii) Singly and doubly magic nuclei.
- (d) List one detector each which can be used for: (i) ionization and track measurements, (ii) time measurements, and (iii) particle identification.
- (e) Explain why it is that only α -particles are emitted by radioactive nuclei while protons and neutrons are not.
- (f) A photon of energy 2.22 MeV can disintegrate the deuteron into its components. Given that the masses of the neutron and proton are 1.008665 u and 1.007277 u respectively, what is the mass of the deuteron?
- (g) What are the informations one can obtain from electron-nucleus scattering experiments?

Question 2.

- (a) The radius of a nucleus is $R = r_0 A^{1/3}$ with $r_0 = 1.2$ fm. Using Heisenberg's relations, estimate the mean kinetic energy and momentum of a nucleon inside a nucleus. [5]
- (b) Write down the Bethe-Weizsäcker mass formula and explain the various terms that contribute to it. [4]
- (c) Describe in brief the Fermi gas model of the nucleus. Show that the Fermi energy is independent of the number of nucleons in the nucleus for constant nuclear density. [5]

Question 3.

- (a) With the help of a neat diagram, describe the characteristics of the mode of operation of gas-filled detectors. [5]

- (b) List some of the important features of a scintillating signal and discuss at least two important features with suitable sketches. [4]

- (c) A NaI(Tl) detector in the shape of a 5 cm cube is bombarded by a beam of 2.8 MeV gamma-radiation normal to one face of the cube. Calculate:

- (i) What fraction of the gamma radiation is detected?
 (ii) What fraction of the detected gamma appears in the photopeak, Compton distribution, and pair peaks?

Assume that there is no re-absorption of Compton gamma or annihilation quanta. (Given: attenuation coefficient μ for 2.8 MeV gammas is 0.135 cm^{-1} , photoelectric coefficient $\mu_p = 2.5 \times 10^{-3} \text{ cm}^{-1}$, Compton coefficient $\mu_{\text{compton}} = 0.113 \text{ cm}^{-1}$, and pair production coefficient $\mu_{\text{pair}} = 0.020 \text{ cm}^{-1}$). [5]

Question 4.

- (a) What are the informations we can obtain from the nucleon-nucleon ($n-p$ and $p-p$) scattering experiments? [6]
 (b) What do you know about the basic properties of the nucleus? Explain any two. Show that the density of nuclear matter is $\approx 10^{17} \text{ kg/m}^3$ (or $\approx 10^5 \text{ tons/mm}^3$). (Assume that the mass of the proton is $1.67 \times 10^{-27} \text{ kg}$ and that its spherical shape has a radius $R = r_0 A^{1/3}$, where $r_0 = 1.2 \times 10^{-13} \text{ cm}$). [7]

OR

- (a) Define the Q -value of alpha decay. Write down the important features of alpha decay. What causes alpha decay? Explain it. [6]
 (b) Calculate the Q -value, kinetic energy T_α , and the Coulomb barrier V_c for the primary branch of the alpha decay of ^{212}Po to the ground state of ^{208}Pb . Here, the mass excess of alpha is 2.4249 MeV, and excess masses of the daughter and parent are -21.759 MeV and -10.381 MeV , respectively. [7]

Question 5.

- (a) What are the properties of the ground state of the deuteron? What are the informations about the nucleon-nucleon potential obtained from the experimental data on the deuteron? [9]
 (b) Make a neat sketch of the binding energy curve. Explain the basis for fission and fusion of nuclei. [5]

OR

- (a) What are elementary particles and how are they classified? Describe the different types of interactions that can occur between the elementary particles. [7]

- (b) Provide the names, masses, and charges of six quarks discovered so far. What do you understand by the colour degree of freedom? What is the need for introducing the colour degree of freedom for the quarks? [7]